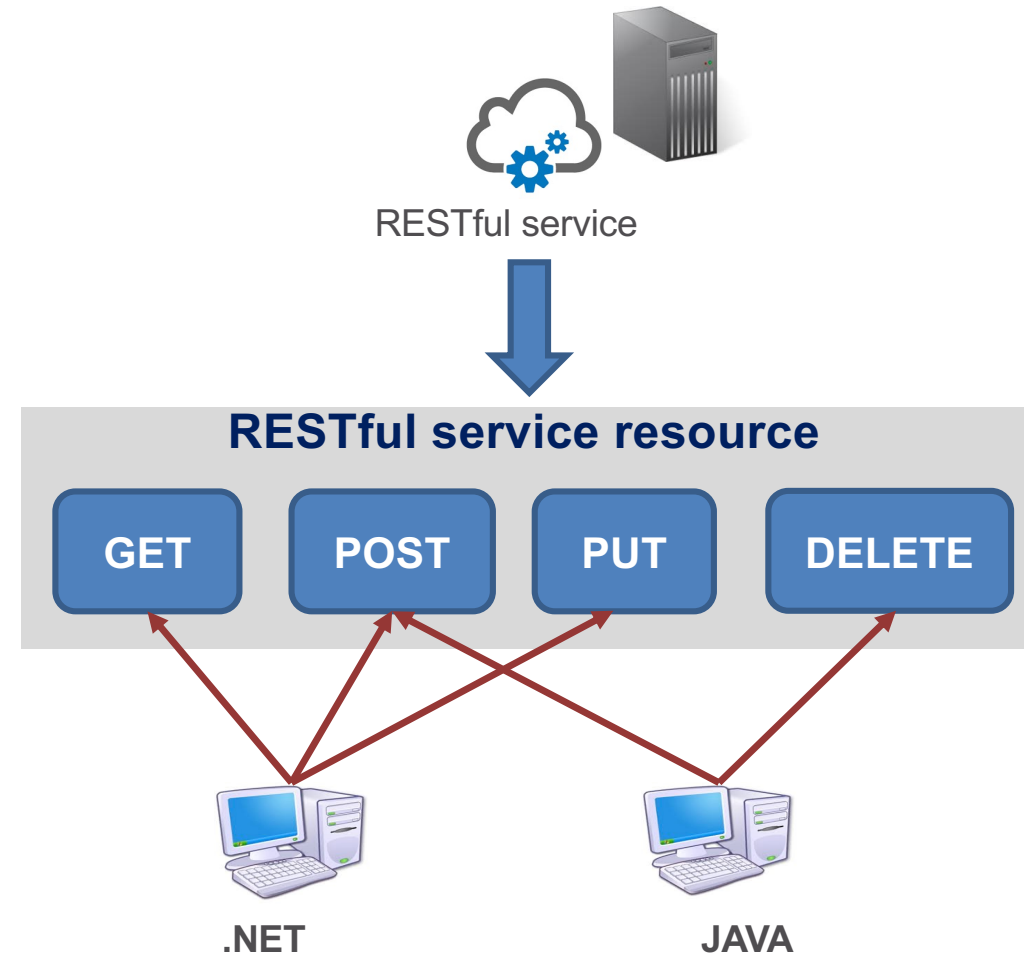


# RESTful Web services

# RESTful Web services

- RESTful architecture relies on RESTful services
- RESTful services are exposed as resources
- The methods of services are exposed as HTTP operations

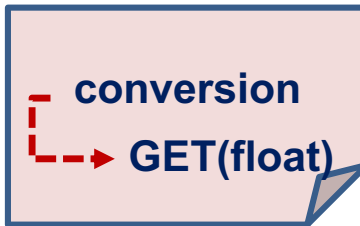


# Example

GET  
<http://compagny.conv/conversion/1500.98>



Client



Web service

ConversionWS

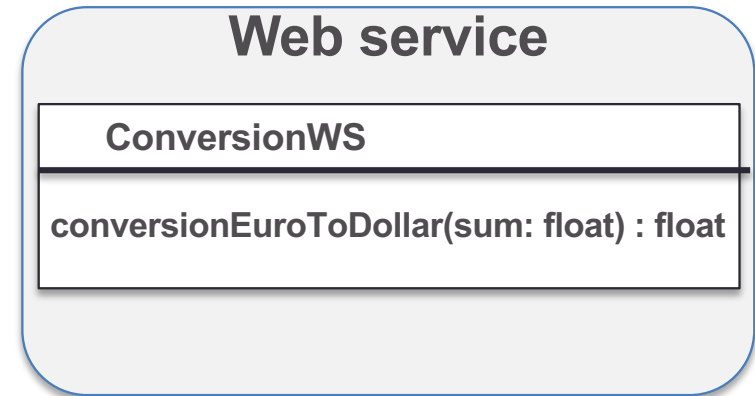
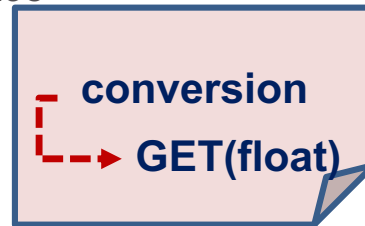
conversionEuroToDollar(sum: float) : float

# Example

GET http://compagny.conv/conversion/1500.98



Client



**GET** http://compagny.conv.fr/conversion/1500.98

The GET  
operation

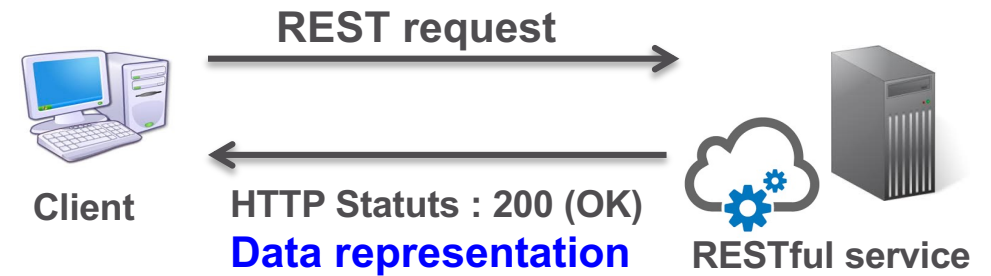
Deployment address

The resource  
name

The parameter  
value

# Data format

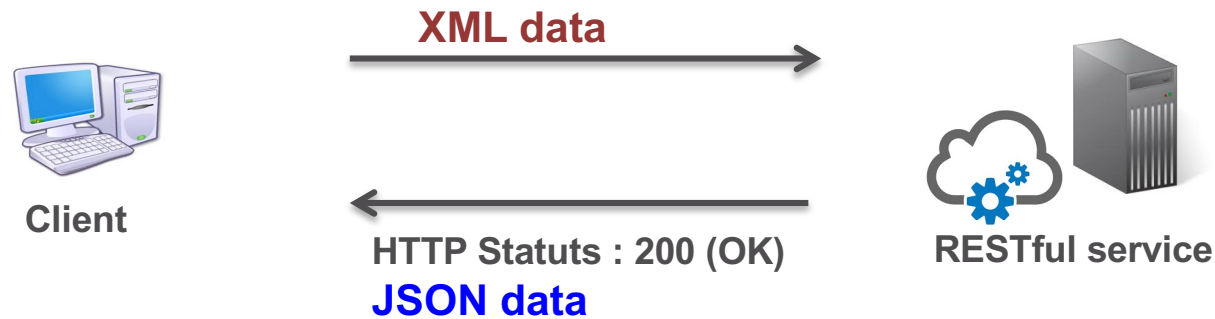
- **XML** (eXtensible Markup Language)



- **JSON** (JavaScript Object Notation)
- **(X)HTML** (eXtensible HyperText Markup Language)

# Data format

- The input data (request) format and output data (response) format **can be different**



# JSON vs XML

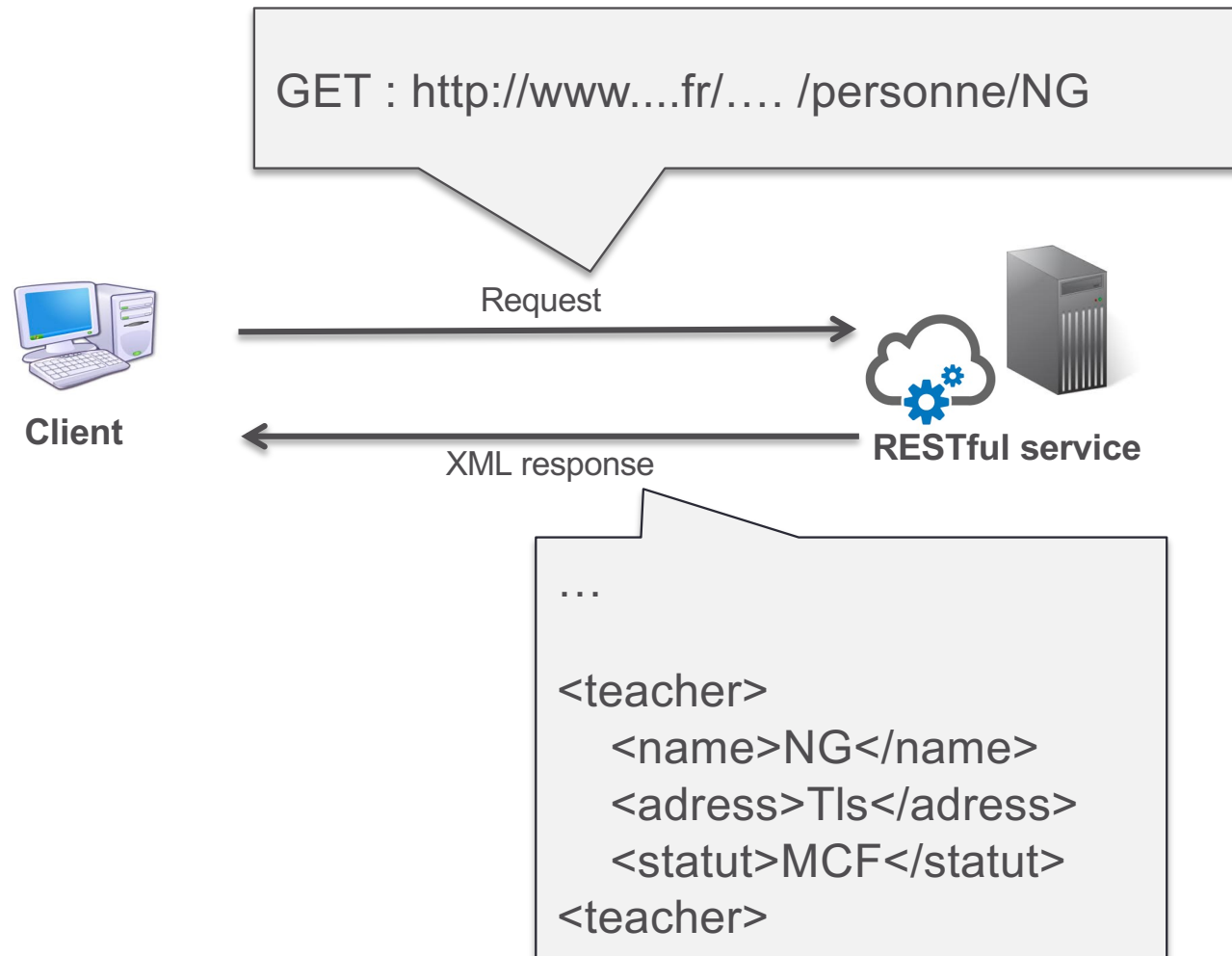
```
<teacher>
  <name>NG</name>
  <address>Tls</adress>
  <status>MCF</status>
</teacher>
```

**XML**

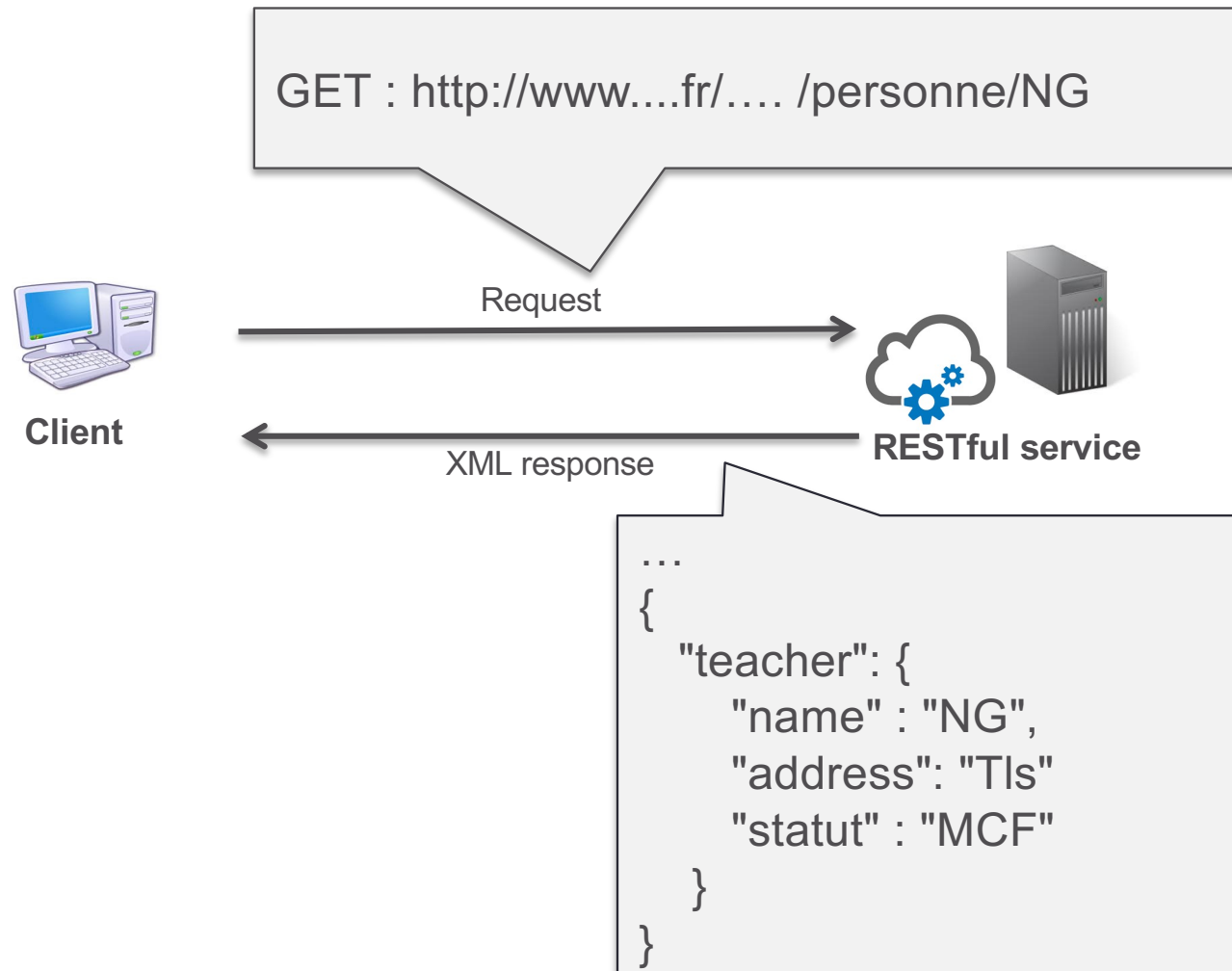
```
{
  "teacher": {
    "name" : "NG",
    "address": "Tls"
    "status" : "MCF"
  }
}
```

**JSON**

# Example of XML response



# Example of JSON response



# SOAP vs REST exchanges

## Web service

ConversionWS

conversionEuroToDollar(sum: float) : float

```
<Envelope>
  <Header/>
  <Body>
    <conversionEuroToDollar>
      <sum>100.25</sum>
    </conversionEuroToDollar>
  </Body>
</Envelope>
```



Client



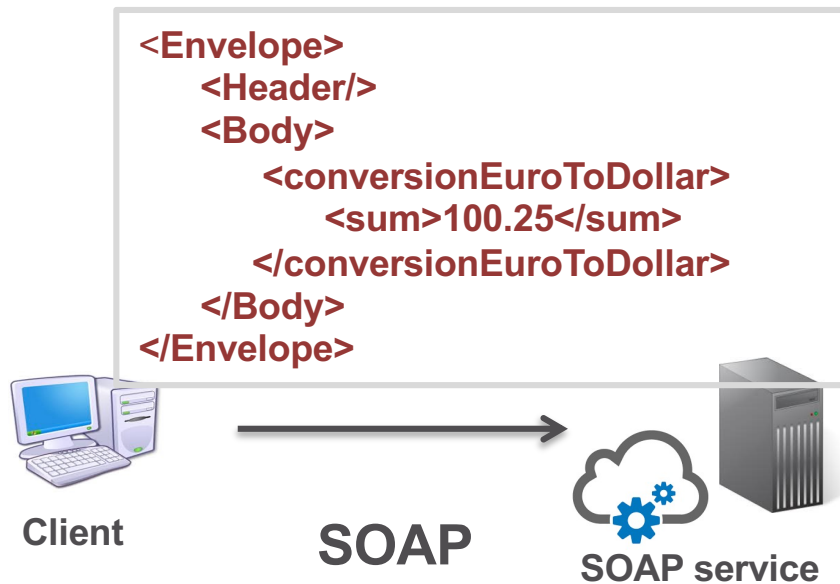
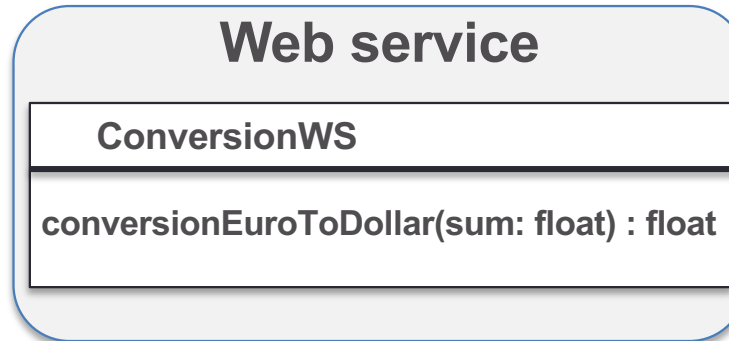
SOAP



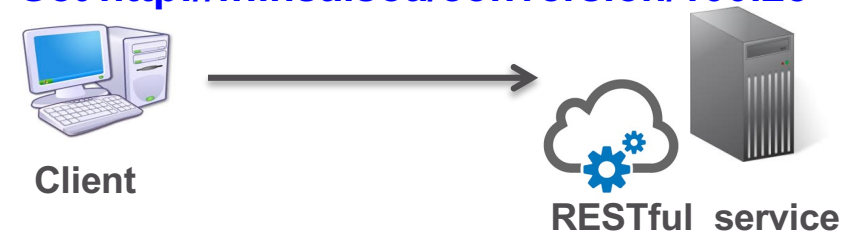
SOAP service



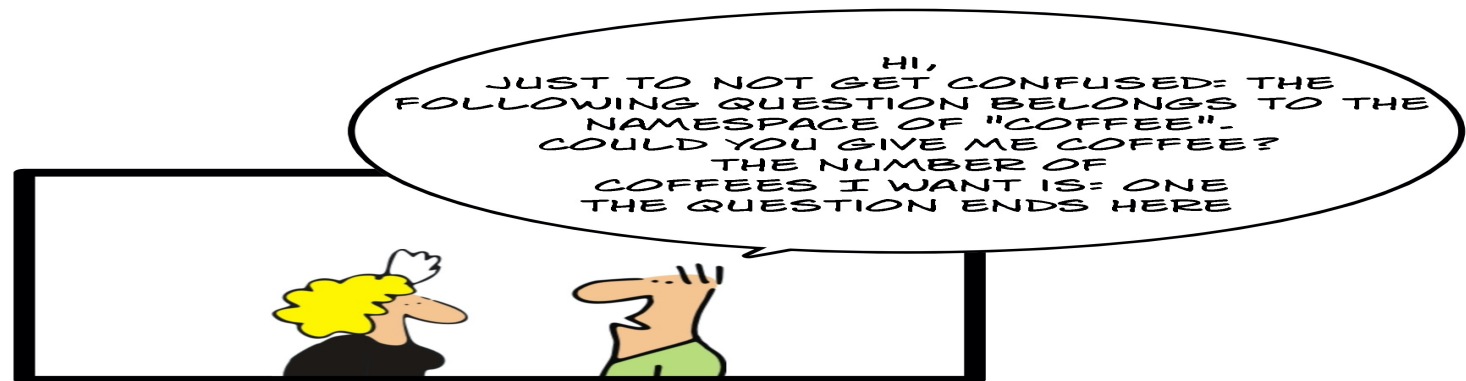
# SOAP vs REST exchanges



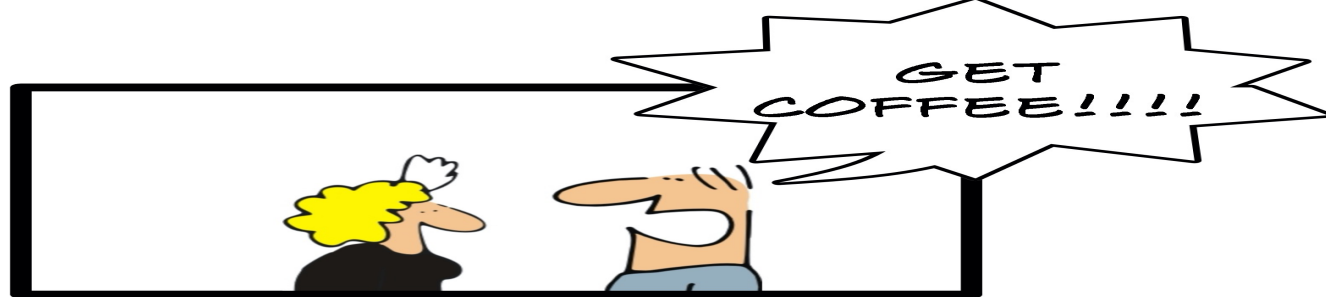
Get <http://fr.insa.soa/conversion/100.25>



# SERVICE CALLING MADE EASY



geek & police



RULE 1: SOAP IS MUCH MORE  
POLITE THAN REST

# RESTfull architecture

- Resource oriented architecture
- RESTFul Web services
- HTTP protocol
- Data exchange format

